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Audit Session

ID: 946C74

OVERVIEW

METRICS

FEATURE IMPORTANCE

GEMINI REPORT

REMEDIATION

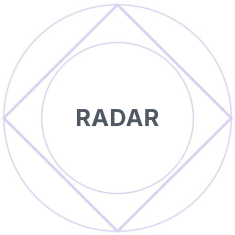
RISK ASSESSMENT

HIGH SEVERITY

4 of 5 metrics failed.

Audit Summary

4 of 5 core fairness metrics failed. age and sex and race are significant predictors of outcome.



FAIRNESS METRICS APR 25, 2026

Demographic Parity Violation FAIL

Measures difference in selection rates across demographic groups.

1.000 vs < 0.1 target

31 66 48

Equalized Odds Violation FAIL

Measures difference in true positive and false positive rates.

1.0

Export Comprehensive Report

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Predictive Parity Violation

FAIL

Measures difference in precision across groups.

1.000 vs < 0.1 target

31 66 48

Disparate Impact

FAIL

Ratio of selection rates (min/max). Below 0.8 typically signals disparate impact.

0.000 vs > 0.8 target

31 66 48

Individual Fairness Violation

AMBER

Proxy based on k-NN consistency. Checks if similar inputs receive similar predictions.

0.126 vs < 0.1 target

31 66 48

Outcome Rate by Protected Group

Outcome Rate by Group

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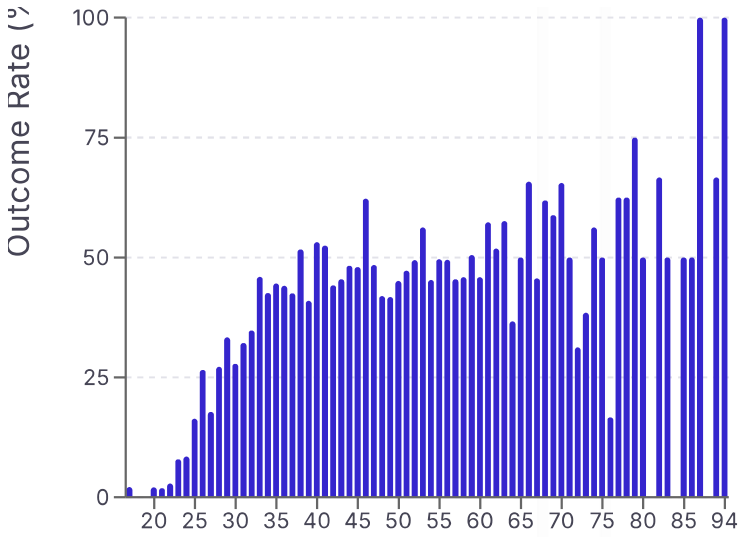

Positive outcome rates per group in the age protected attribute

What's Driving the Model?

WORK_HOURS_PER_WEEK	+0.70
EDUCATION_LEVEL	+0.43
RELATIONSHIP	+0.38
MARITAL_STATUS	+0.36
OCCUPATION_CODE	+0.33
PLACE_OF_BIRTH	+0.17
CLASS_OF_WORKER	+0.11

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REPORT

Audit Findings

SUMMARY

Our analysis of the AI model designed to predict income for hiring purposes has revealed critical and severe issues of unfairness. In its current state, the model is highly discriminatory against certain groups of people based on their race, sex, and age. The problem is so severe that for at least one demographic group, the model predicts a 0% chance of success, effectively disqualifying every single person from that group. This is indicated by a 'Disparate Impact' score of 0.0, where a fair score would be 0.8 or higher. The root cause is not that the model was explicitly told to use race or sex, but that it learned to rely heavily on factors like 'marital status' and 'relationship status'. Because these factors are often linked to a person's gender and age in historical data, the model has inadvertently learned to replicate and amplify historical societal biases. For example, it may have learned to favor married men simply because they have historically held higher-paying jobs. Using this model would expose our organization to significant legal, ethical, and reputational risks. We must immediately halt any use of this model and begin a comprehensive process to rebuild it with fairness as a primary goal.

WHO IS AFFECTED

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Unspecified gender groups (likely 'Female' given the strong influence of proxy features like 'relationship' and 'marital_status')

Unspecified age groups

ROOT CAUSE

The extreme fairness violations stem from indirect discrimination, where the model heavily relies on features that are strong proxies for the protected attributes of race, sex, and age. The top SHAP features 'relationship' (0.3848) and 'marital_status' (0.3603) are historically correlated with gender and age, embedding societal biases into the model's predictions. The Disparate Impact score of 0.0 indicates the model has learned a rule that effectively filters out an entire unprivileged subgroup, likely a severely underrepresented racial group, from receiving a positive prediction.

SUGGESTED:

How can I mitigate race proxy bias?

Which group is most affected by this model?

What are the top remediation steps I should take?

Ask Gemini about this au

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01 Remove strong proxy features: Immediately remove 'relationship' and 'marital_status' from the feature set and retrain the model. Evaluate 'occupation_code' for potential bias and consider grouping it into broader categories.

02 Implement data-level re-balancing: Use preprocessing techniques like SMOTE (Synthetic Minority Over-sampling Technique) or sample re-weighting to address the class imbalance and severe underrepresentation of minority racial groups in the training data.

03 Apply fairness-aware modeling: Retrain the model using an in-processing fairness algorithm (e.g., Exponentiated Gradient, Adversarial Debiasing) to enforce fairness constraints like Demographic Parity or Equalized Odds during the training process.

04 Establish group-specific thresholds (post-processing): As a more immediate fix,

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equalize the selection rates or error rates post-prediction. This should be a temporary measure while more fundamental changes are made.

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